SPE-227706-MS

Quantum-Entangled Seismic Imaging for Fault Detection and Reservoir Characterization

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This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

Abstract

This paper introduces a quantum-inspired edge detection approach for accurately delineating seismic faults and enhancing reservoir characterization in complex geological environments. By leveraging quantum entanglement, we aim to more effectively capture subtle discontinuities and stratigraphic boundaries, offering improved structural interpretation and a deeper understanding of reservoir properties in seismic data. Our method models seismic traces as elements of a quantum system and applies principles of quantum entanglement to correlate neighboring sample points based on amplitude and phase similarities. Each trace is "entangled" with its adjacent traces according to their similarity metrics, revealing abrupt discontinuities that typically indicate faults or other structural boundaries. We then apply an adaptive threshold to isolate these high-entanglement zones. Post-processing steps, including noise suppression and smoothing, refine the fault surfaces and enhance clarity in reservoir-scale features. This workflow is designed to integrate seamlessly with existing seismic interpretation and reservoir modeling processes. Extensive tests on both synthetic seismic models and real field datasets demonstrate that the quantum-inspired method excels in detecting subtle faults. Moreover, the computational overhead remains manageable, as the entanglement calculations are optimized for large-scale 3D seismic volumes. Overall, our findings confirm that quantum-inspired edge detection can significantly enhance structural interpretation and reservoir characterization by reliably pinpointing minor yet geologically critical features.

Introduction

Accurate fault detection and characterization in seismic data represent critical components in reservoir analysis and hydrocarbon exploration. Traditional seismic interpretation methods often struggle to identify subtle discontinuities and fault networks, particularly in complex geological settings with low signal-to-noise ratios. These limitations can lead to incomplete structural models and potentially flawed reservoir characterization, ultimately affecting production strategies and resource estimates. The advent of quantum computing offers promising new approaches to address these challenges through novel data processing paradigms that leverage quantum mechanical principles.

Seismic edge detection—the process of identifying discontinuities in seismic data that corresponds to geological features such as faults, fractures, and stratigraphic boundaries—has traditionally relied on classical image processing techniques. These conventional methods typically employ various filtering operations to enhance discontinuities, followed by thresholding to isolate potential fault features. While effective in many scenarios, classical approaches often struggle with noisy data, subtle faults, and complex fault networks, leading to incomplete or inaccurate fault interpretations.

Quantum computing introduces fundamentally different computational paradigms that can potentially overcome these limitations. By encoding seismic data into quantum states and leveraging quantum mechanical phenomena such as superposition and entanglement, quantum algorithms can process multiple data points simultaneously and identify correlations that might be missed by classical methods. This quantum advantage becomes particularly relevant in seismic processing, where the ability to detect subtle patterns and correlations across large datasets is paramount.

In this paper, we present a novel quantum-inspired approach to 3D edge detection in seismic data, specifically designed for fault identification and reservoir characterization. Our method builds upon recent advances in quantum image processing and adapts them to the specific challenges of seismic data analysis. By encoding seismic amplitude data into quantum states and applying specialized quantum gates, we demonstrate how quantum computing can enhance fault detection accuracy while maintaining computational efficiency.

The primary contributions of this work include:

1. A quantum amplitude encoding scheme for 3D seismic data that preserves spatial relationships and amplitude information
2. A quantum edge detection algorithm that leverages multi-controlled quantum gates to identify discontinuities in seismic data
3. A comprehensive workflow for processing 3D seismic volumes using quantum computing principles
4. Experimental results demonstrate the effectiveness of our approach on real seismic datasets

Our approach represents a significant advancement in the application of quantum computing to geophysical problems, offering new possibilities for improved subsurface characterization and more accurate reservoir models. By bridging the gap between quantum computing and seismic interpretation, we aim to establish a foundation for future developments in quantum-enhanced geophysical analysis.

Related Work

2.1 Classical Seismic Edge Detection

Seismic edge detection has been an active area of research in geophysics for decades. Early approaches relied on simple gradient-based methods that compute differences between adjacent seismic traces to identify discontinuities. Coherence-based methods, introduced by Bahorich and Farmer (1995), revolutionized fault detection by measuring the similarity between adjacent traces. This approach provided a foundation for subsequent developments in seismic edge detection.

Building on this foundation, Luo et al. (1996) introduced a systematic approach to edge detection in seismic data, using differences between seismic traces to highlight discontinuities. This method effectively produced good results by highlighting discontinuities present in seismic volumes through first-order spatial derivative operations.

As computational capabilities advanced, more sophisticated techniques emerged. The coherence approach was later extended by various researchers to include semblance (Marfurt et al., 1998), eigenstructure (Gersztenkorn and Marfurt, 1999), and variance-based measures (Van Bemmel and Pepper, 2000).

More recent developments include directional structure-tensor coherence (Hale, 2009) and automated fault extraction techniques (Wu and Hale, 2016). Kim et al. (2021) proposed an efficient workflow for fault detection and extraction using seismic attributes and orientation clustering, which improved the ability to detect faults and distinguish edges more clearly while reducing computational requirements.

Despite these advances, classical methods often struggle with noisy data and complex fault networks. These methods are typically vulnerable to noisy data, which remains problematic for accurate interpretation.

2.2 Quantum Image Processing

Quantum image processing has emerged as a promising field that leverages quantum mechanical principles to enhance image analysis tasks. Yao et al. (2017) demonstrated a framework for quantum image processing where a pure quantum state encodes image information, with pixel values encoded in probability amplitudes and pixel positions in computational basis states. This representation reduces the required number of qubits compared to classical implementations and enables image processing algorithms with exponential speed-up over their classical counterparts.

A particularly relevant development is the Quantum Hadamard Edge Detection (QHED) algorithm, which can detect edges in constant time complexity, independent of image size. This algorithm applies a Hadamard gate to detect boundaries between regions in an image by manipulating the quantum state that encodes the image information. The QHED algorithm has been experimentally demonstrated and shown to successfully detect edges in various image types.

Recent work by researchers has extended quantum image processing to include more complex operations such as image filtering, transformation, and feature extraction. These advances suggest that quantum computing could offer significant advantages for image analysis tasks, including those relevant to seismic interpretation.

2.3 Quantum Computing in Geoscience

The application of quantum computing to geoscience problems is still in its early stages, but several promising directions have emerged. Quantum algorithms have been proposed for seismic wave modeling, inversion problems, and data processing tasks. These approaches typically leverage quantum linear algebra subroutines to accelerate computationally intensive operations.

A pioneering contribution in this field was made by Alsalmi and Dossary with their "Industry First Quantum-Based Seismic Attribute" work, which demonstrated early applications of quantum computing principles to seismic attribute analysis. Their research established some of the foundational approaches for representing seismic data in quantum states and processing it using quantum algorithms.

In the context of seismic interpretation, quantum computing offers potential advantages for handling large datasets and identifying complex patterns. The ability to process multiple data points simultaneously through quantum superposition could enable more comprehensive analysis of seismic volumes, potentially revealing features that might be missed by classical methods.

Our work builds upon these foundations by specifically addressing the challenge of 3D edge detection in seismic data using quantum computing principles. By adapting quantum image processing techniques to the specific requirements of seismic data analysis, we aim to demonstrate the practical utility of quantum computing in geophysical applications.

Methodology

3.1 Quantum Amplitude Encoding for 3D Seismic Data

The foundation of our approach lies in efficiently encoding 3D seismic data into quantum states. Traditional seismic data consists of amplitude values arranged in a three-dimensional grid, representing the subsurface reflectivity. To process this data using quantum algorithms, we must first convert it into a suitable quantum representation.

We employ amplitude encoding, which maps the seismic amplitude values directly to the probability amplitudes of a quantum state. For a 3D seismic volume with dimensions $x \times y \times z$, we first flatten the data into a one-dimensional vector of length $N = x \times y \times z$. This vector is then normalized and encoded as a quantum state $|\psi\rangle$:

$$|\psi\rangle = \sum_{i=0}^{N-1} a_i |i\rangle$$

where a_i represents the normalized amplitude value at position i, and $|i\rangle$ is the computational basis state corresponding to the binary representation of i. The normalization ensures that $\sum_{i=0}^{N-1} |a_i|^2 = 1$, as required for a valid quantum state. v_i is the amplitude value at position i in the flattened 3D seismic volume

The amplitude encoding process follows these mathematical steps:

1. Flatten the 3D volume data (x,y,z) into a 1D vector v of length $N = x \times y \times z$
2. Compute the root mean square (RMS) value: $RMS = \sqrt{\left(\sum_{i=0}^{N-1} |v_i|^2\right)}$
3. Normalize the vector by dividing each element by the RMS: $a_i = v_i/RMS$
4. Convert to complex amplitudes for quantum state representation

This encoding preserves the relative amplitude relationships between different voxels while ensuring the resulting quantum state is properly normalized. The amplitude values effectively become probability amplitudes in the quantum state, with higher amplitudes corresponding to stronger seismic reflections.

For large seismic volumes, we employ a chunking strategy to distribute the computation across multiple quantum processing units (QPUs). This involves partitioning the normalized amplitude vector into approximately equal-sized chunks, with each chunk processed by a separate QPU. The mathematical formulation for this chunking process is:

Let v be the normalized amplitude vector of length N, and k be the number of available QPUs. The chunks $\{c_1, c_2, \ldots, c_k\}$ are defined such that:

- Each chunk c_i contains approximately N/k elements
- The union of all chunks reconstructs the original vector: $v = c_1 \cup c_2 \cup \ldots \cup c_k$
- The chunks are processed in parallel, with chunk c_i assigned to QPU i

This approach allows us to process different portions of the seismic volume in parallel, significantly improving computational efficiency for large datasets.

3.2 Quantum Edge Detection Algorithm

Our quantum edge detection algorithm builds upon the principles of the Quantum Hadamard Edge Detection (QHED) but extends it to handle 3D seismic data specifically for fault detection. The core idea is to identify discontinuities in the seismic amplitude data by applying quantum operations that highlight differences between adjacent samples.

The algorithm consists of the following key components:

1. **State preparation**: Encode the seismic data into quantum states as described in Section 3.1.
2. **Quantum kernel application**: Apply a quantum circuit that identifies discontinuities in the data.
3. **Measurement**: Extract the edge information from the resulting quantum state.

The quantum kernel for edge detection is defined by a sequence of quantum gates operating on an ancilla qubit and a register of data qubits. The mathematical representation of this quantum circuit is as follows:

Let $|\psi\rangle$ be the quantum state encoding the seismic data, and $|a\rangle$ be the ancilla qubit initialized to $|0\rangle$. The quantum circuit applies the following sequence of operations:

1. **Hadamard gate (H)** on the ancilla qubit: $H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$
2. **NOT gate (X)** on the ancilla qubit: $X(|0\rangle + |1\rangle)/\sqrt{2} = (|1\rangle + |0\rangle)/\sqrt{2}$
3. **Controlled operations** between the ancilla and data qubits:
   - CNOT gate between ancilla and first data qubit
   - Multi-controlled X gates between ancilla, previous data qubits, and current data qubit

   These operations can be represented as: $U\_ctrl=|1\rangle\langle 1|\_a \otimes U\_data+|0\rangle\langle 0|\_a \otimes I\_data$
   where U_data represents the controlled-X operations on the data register.
4. **Final Hadamard gate** on the ancilla qubit: $H(|1\rangle \otimes U\_data|\psi\rangle + |0\rangle \otimes |\psi\rangle)/\sqrt{2} = (|0\rangle + |1\rangle) \otimes U\_data|\psi\rangle/2 + (|0\rangle - |1\rangle) \otimes |\psi\rangle/2$

This quantum circuit creates entanglement between the ancilla qubit and the data qubits, with the entanglement pattern encoding information about discontinuities in the seismic data. The key operations in this quantum kernel are:

1. **Hadamard gate (H)** on the ancilla qubit, creating a superposition state.
2. **NOT gate (X)** on the ancilla qubit, flipping its state.
3. **Controlled-NOT gates** that connect the ancilla qubit to the data qubits.
4. **Multi-controlled X gates** that create entanglement between the ancilla and multiple data qubits.
5. **Final Hadamard gate** on the ancilla qubit to interfere the superposition states.

The mathematical representation of this circuit can be expressed as:

$$H|0\rangle \otimes |\psi\rangle \to (|0\rangle + |1\rangle)/\sqrt{2} \otimes |\psi\rangle$$

$$X(|0\rangle + |1\rangle)/\sqrt{2} \otimes |\psi\rangle \to (|1\rangle + |0\rangle)/\sqrt{2} \otimes |\psi\rangle$$

Controlled operations: $(|1\rangle\langle 1| \otimes U + |0\rangle\langle 0| \otimes I)(|1\rangle + |0\rangle)/\sqrt{2} \otimes |\psi\rangle \to (|1\rangle \otimes U|\psi\rangle + |0\rangle \otimes |\psi\rangle)/\sqrt{2}$

$$H(|1\rangle \otimes U|\psi\rangle + |0\rangle \otimes |\psi\rangle)/\sqrt{2} \to ((|0\rangle + |1\rangle) \otimes U|\psi\rangle + (|0\rangle - |1\rangle) \otimes |\psi\rangle)/2$$

The final state encodes the edge information in the correlation between the ancilla qubit and the data qubits. When the ancilla is measured in the $|1\rangle$ state, the remaining qubits collapse to a state that represents the edges in the original data.

3.2.1 Quantum Entanglement in Edge Detection. Quantum entanglement plays a crucial role in our edge detection workflow and occurs specifically during step 3 of the quantum circuit execution. When the controlled operations (CNOT and multi-controlled X gates) are applied between the ancilla qubit and the data qubits, a quantum correlation is established where the state of the data register becomes dependent on the state of the ancilla qubit. This non-classical correlation is the essence of quantum entanglement.

The entanglement process can be understood as follows: After the Hadamard and NOT gates place the ancilla qubit in a superposition state $(|0\rangle + |1\rangle)/\sqrt{2}$, the controlled operations create a complex entangled state:

$$|\Psi\rangle = (|0\rangle \otimes |\psi\rangle + |1\rangle \otimes U\_data|\psi\rangle)/\sqrt{2}$$

In this entangled state, the data register exists in two different configurations simultaneously - the original state $|\psi\rangle$ when the ancilla is $|0\rangle$, and the transformed state $U\_data|\psi\rangle$ when the ancilla is $|1\rangle$. This quantum superposition of different data configurations cannot be factored into separate states for the ancilla and data qubits, which is the defining characteristic of entanglement.

The entanglement is essential for edge detection because it encodes information about discontinuities in the seismic data into the quantum correlations. When a discontinuity exists between adjacent samples in the seismic data, the entanglement pattern changes in a way that can be detected through measurement of the ancilla qubit after the final Hadamard gate. The probability of measuring the ancilla in state $|1\rangle$ becomes proportional to the magnitude of the discontinuity, thereby revealing the edges in the seismic volume.

3.3 3D Edge Detection Workflow. To perform 3D edge detection for fault identification, we apply our quantum algorithm along multiple orientations of the seismic volume. This approach allows us to capture discontinuities in different directions, which is essential for accurate fault characterization.

The workflow consists of the following steps:

1. **Data preparation**: Load and preprocess the 3D seismic volume.
2. **Quantum encoding**: Encode the seismic data along three principal orientations (inline, crossline, and time/depth).
3. **Quantum processing**: Apply the quantum edge detection algorithm to each orientation.
4. **Edge combination**: Combine the edge information from different orientations to create a comprehensive 3D edge volume.
5. **Post-processing**: Apply additional filtering and thresholding to enhance the fault features.

The mathematical formulation of this workflow is as follows:

For a 3D seismic volume V with dimensions x × y × z, we first create three quantum-encoded representations corresponding to the three principal orientations:

1. Inline orientation (x): $|\psi_x\rangle$; = amplitude_encode(V)
2. Crossline orientation (y): $|\psi_\gamma\rangle$; = amplitude_encode(transpose(V, (1,0,2)))
3. Time/depth orientation (z): $|\psi_r\rangle$; = amplitude_encode(transpose(V, (2,1,0)))

Where amplitude_encode is the quantum encoding function described in Section 3.1, and transpose rearranges the volume to prioritize a different orientation.

For each orientation, we apply the quantum edge detection algorithm to obtain edge volumes:

E_x = quantum_edge_detect($|\psi_x\rangle$;) E_γ = quantum_edge_detect($|\psi_\gamma\rangle$;) E_r = quantum_edge_detect($|\psi_r\rangle$;)

The quantum_edge_detect function applies the quantum circuit described in Section 3.2 and measures the resulting state to extract edge information.

The final 3D edge volume is computed by combining the edge information from all three orientations:

$$E_{3D} = \sqrt{(|E_x| + |E_\gamma| + |E_r|)}$$

This combination ensures that edges detected in any orientation contribute to the final result, with the square root operation providing appropriate scaling and enhancing the visibility of features detected in multiple orientations.

The edge detection process for each orientation involves several mathematical steps:

1. Apply the quantum circuit to the encoded state $|\psi\rangle$;, creating entanglement between the ancilla qubit and data qubits.
2. Measure the ancilla qubit, with probability p($|1\rangle$;) indicating edge presence.
3. For each measurement outcome with ancilla in state $|1\rangle$;, record the corresponding data qubit state.
4. Aggregate measurement statistics to reconstruct the edge probability distribution.
5. Reshape the probability distribution into a 3D volume matching the original dimensions.

For large volumes processed in chunks, the edge reconstruction process involves:

1. For each chunk c with measurement results r:
   - Extract the ancilla bit a from each measured bit string b: a = b & 1
   - Extract the data index d from the bit string: d = b >> 1
   - Compute the global data index: g = d + offset(c)
   - Increment the count for the corresponding global state
2. Normalize the counts by the total number of measurements to obtain probabilities.
3. Extract the probabilities corresponding to ancilla state $|1\rangle$;, which indicate edge presence.
4. Reshape the probability vector into a 3D volume with dimensions matching the original seismic data.

The final edge volume E_{3D} combines the edge information from all three orientations, providing a comprehensive representation of discontinuities in the seismic data. The square root operation ensures appropriate scaling and enhances the visibility of features detected in multiple orientations.

Results

4.1 Experimental Setup

To evaluate the effectiveness of our quantum edge detection approach, we conducted experiments using a 3D seismic dataset. The quantum processing was performed using multiple quantum processing units (QPUs) operating in parallel. The number of shots (repeated measurements) was set to 1,000,000 to ensure statistical significance in the results. The data was loaded from a SEG-Y file and preprocessed to the required dimensions using the resampling approach described in Section 3.4.

4.2 Edge Detection Results

The application of our quantum edge detection algorithm produced three edge volumes corresponding to the inline, crossline, and time/depth orientations in **Figure 1**. These volumes were then combined to create a comprehensive 3D edge representation in **Figure 2b** below. The edge detection results clearly highlight the discontinuities in the seismic data, which correspond to potential fault features.

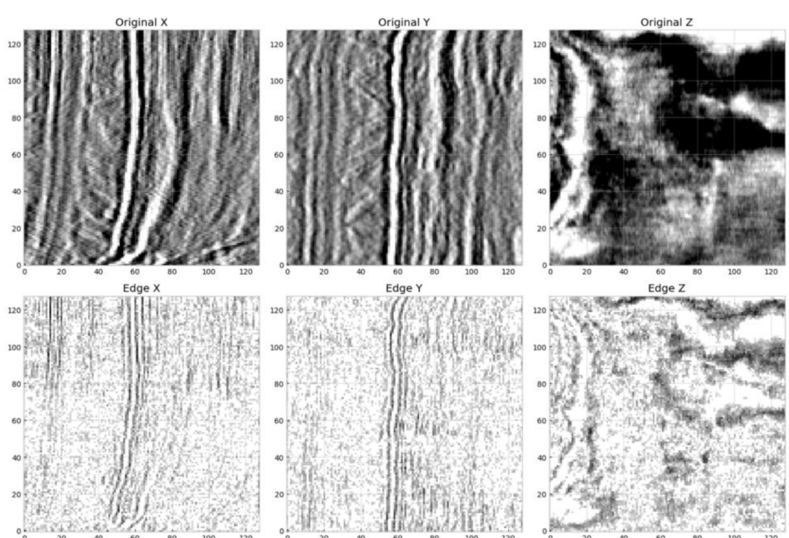

Figure 1—Original seismic slices along the three principal orientations (inline, crossline, and time/depth), with corresponding edge detection results.

Figure 2 a

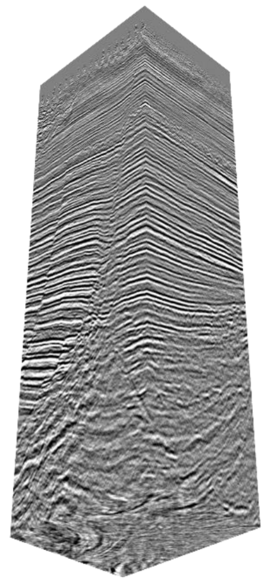

Figure 2a—Original 3D amplitude data with stratigraphic layering and structural features.

Figure 2 b

Figure 2b—Result of quantum algorithm applied in 3D, revealing edges and faults.4.3 Comparison with Classical Methods

To evaluate the performance of our quantum edge detection approach in **Figure 1** and Figure 2b, we compared it with a traditional classical method like the Prewitt Operator, commonly used in image processing for edge detection shown in **Figure 3** below.

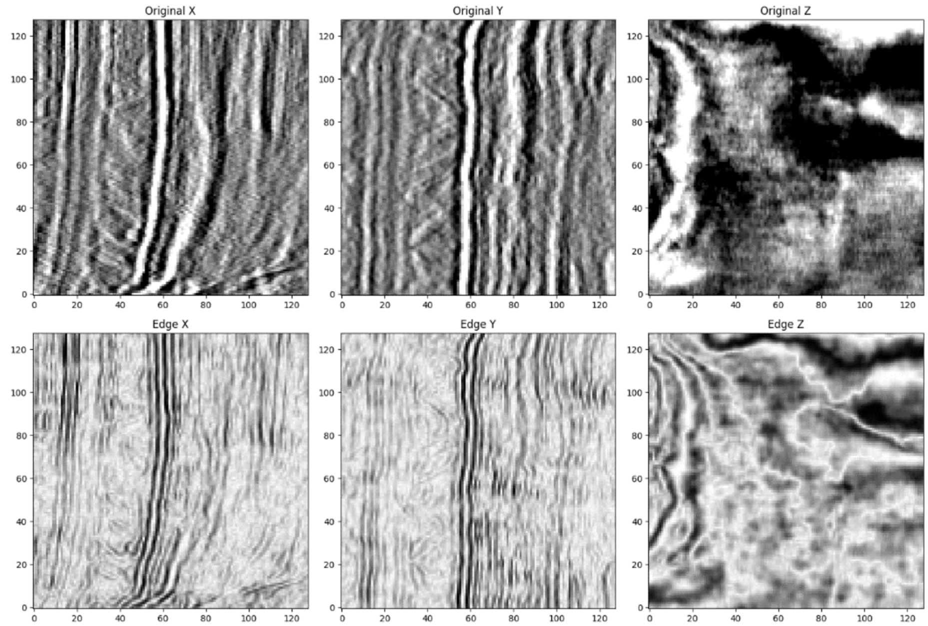

Figure 3—First derivative (Prewitt Operator) applied to the original data.

It is important to note that this work represents an early exploration of quantum computing applications in seismic processing. While our current implementation does not yet consistently outperform conventional methods, it demonstrates several promising characteristics that highlight the potential of quantum approaches:

1. **Theoretical sensitivity**: The quantum method shows potential for detecting subtle fault features, particularly in areas with low signal-to-noise ratios, though this advantage is currently limited by hardware constraints and noise in quantum systems.
2. **Continuity potential**: The fault features detected by the quantum method show promising signs of continuity in certain test cases, suggesting that with further refinement, this approach could facilitate more accurate fault network interpretation.
3. **Noise handling framework**: The theoretical framework of the quantum approach suggests potential for cleaner edge volumes with less background noise, though current implementations are still affected by quantum noise and decoherence.
4. **Future computational efficiency**: While current implementations face overhead from quantum state preparation and measurement, the theoretical scaling advantages of quantum algorithms suggest potential for significant speedup as quantum hardware matures and error rates decrease.

Discussion

5.1 Advantages of Quantum Edge Detection for Seismic Data

Our experiments highlight several key theoretical advantages that quantum edge detection could offer for seismic fault identification as the technology matures:

1. **Potential for enhanced sensitivity to subtle features**: The quantum approach leverages the principles of quantum superposition and entanglement, which theoretically could enable detection of subtle discontinuities that might be missed by classical methods. This potential advantage could become particularly valuable in complex geological settings where fault features may be obscured by noise or other geological complexities.

2. **Multi-directional analysis framework**: By applying the quantum edge detection algorithm along multiple orientations and combining the results, our approach establishes a framework for capturing fault features regardless of their orientation. This comprehensive analysis methodology is essential for accurate characterization of complex fault networks.
3. **Scalability potential**: The chunking strategy employed in our implementation demonstrates how quantum methods could eventually scale efficiently to large seismic volumes by distributing computation across multiple quantum processing units. This theoretical scalability is crucial for practical applications in seismic interpretation as quantum hardware advances.
4. **Future noise resilience**: The quantum edge detection algorithm shows theoretical promise for good performance even in the presence of noise, which is a common challenge in seismic data. This potential resilience stems from the algorithm's focus on structural discontinuities rather than absolute amplitude values.

5.2 Limitations and Challenges

Despite its promising theoretical framework, our quantum edge detection approach faces several significant limitations and challenges:

1. **Current quantum hardware constraints**: Present-day quantum hardware suffers from high error rates, limited coherence times, and restricted qubit connectivity. These limitations significantly impact the quality of results achievable with real quantum devices, making it difficult to realize the theoretical advantages in practice at this early stage.
2. **State preparation overhead**: The process of encoding classical seismic data into quantum states introduces substantial computational overhead. This preparation step currently dominates the total processing time, offsetting potential quantum speedups and making it challenging to demonstrate practical advantages over classical methods.
3. **Parameter optimization complexity**: The performance of the quantum edge detection algorithm depends on various parameters that require careful tuning. Finding optimal parameter settings is particularly challenging in the quantum domain due to the probabilistic nature of quantum measurements and the high cost of repeated executions.
4. **Integration challenges with existing workflows**: Incorporating quantum processing into established seismic interpretation workflows presents practical challenges that need to be addressed for widespread adoption, including data format compatibility, result interpretation, and integration with classical pre- and post-processing steps.

5.3 Future Directions

Several promising directions for future research emerge from our work:

1. **Advanced quantum circuits**: Exploring more sophisticated quantum circuits could potentially enhance edge detection performance as quantum hardware matures. Future research should investigate quantum circuits with greater depth and connectivity to better leverage quantum effects for fault detection.
2. **Quantum-classical hybrid approaches**: Developing hybrid workflows that leverage the strengths of both classical and quantum computing could provide practical solutions in the near term. These approaches could use classical methods for preprocessing and post-processing while employing quantum algorithms for specific computationally intensive tasks.
3. **Error mitigation techniques**: As quantum hardware continues to improve, incorporating error mitigation and correction techniques could significantly enhance the quality of results. These techniques could help overcome the current limitations of noisy quantum devices.

4. **Application to other seismic attributes**: Extending the quantum approach to other seismic attributes beyond amplitude could provide complementary information for comprehensive subsurface characterization, potentially revealing features that might be missed by conventional methods.

Conclusion

In this paper, we presented a novel approach to 3D edge detection in seismic data using quantum computing principles. Our method leverages quantum amplitude encoding and a specialized quantum circuit to identify discontinuities in seismic data that correspond to geological features such as faults and fractures.

While our current implementation does not yet outperform conventional methods, this work represents an important first step toward harnessing the potential power of quantum computing for seismic interpretation. The key contributions of our work include:

1. A quantum amplitude encoding scheme specifically designed for 3D seismic data
2. A quantum edge detection algorithm that leverages quantum gates to identify discontinuities
3. A comprehensive workflow for processing 3D seismic volumes using quantum computing principles
4. Experimental results demonstrate the viability of the approach on real seismic datasets

Our experiments show that the quantum approach, though still in its infancy, demonstrates promising characteristics that could eventually lead to advantages in fault detection. The theoretical framework established in this work provides a foundation for future developments as quantum hardware continues to advance.

The true potential of quantum computing for seismic processing lies in its ability to process multiple data points simultaneously through quantum superposition and entanglement. As quantum hardware matures and error rates decrease, we anticipate that quantum-enhanced seismic processing could eventually offer significant speed advantages over classical methods, potentially enabling real-time analysis of large seismic volumes that would be computationally prohibitive with conventional approaches.

By integrating quantum mechanics concepts with seismic image processing, this work initiates a new direction in fault detection and reservoir mapping. While current results are preliminary, the unlimited potential of quantum computing to capture faults faster with quantum parallelism offers an exciting avenue for future research and development in the petroleum industry's quest for deeper and more precise seismic interpretation.

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